Security Module

Password & Authentication System

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Project Explanation

This project implements real-world cybersecurity practices including password protection, breach detection, and multi-factor authentication.

- Strong password generation
- Breach detection using API
- Two-Factor Authentication (2FA)
- Swagger testing interface

Architecture

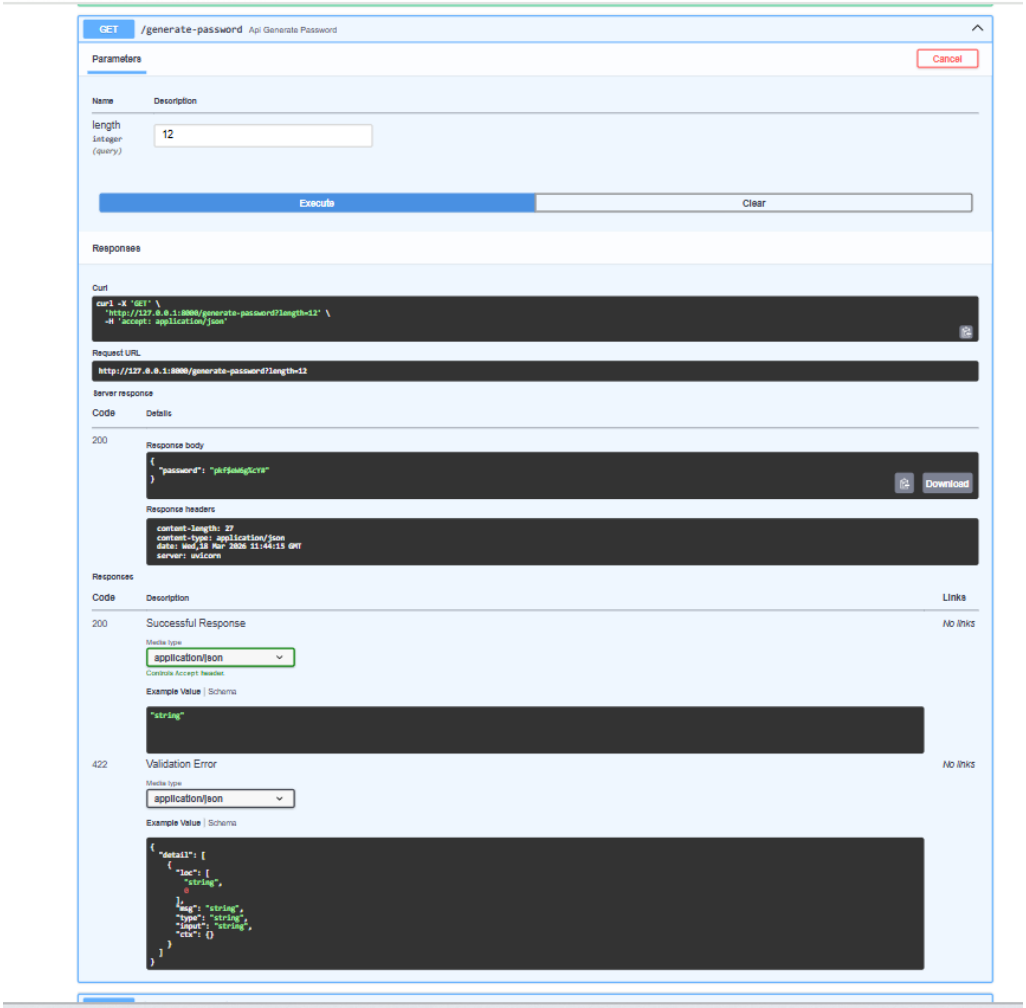
User → Swagger UI → FastAPI → HIBP API
↓
2FA System (pyotp)
↓
QR Code

API Endpoints

Generate Password

`GET /generate-password?length=12`

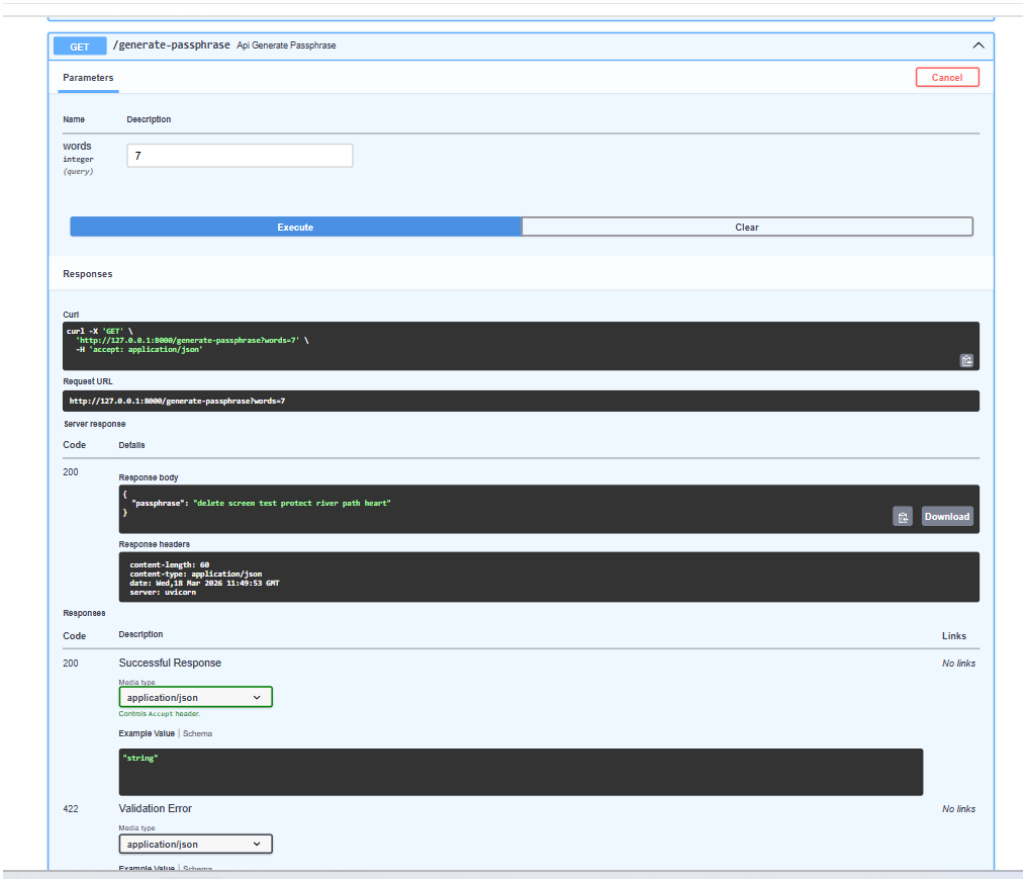
- Generates strong password
- Includes letters, numbers, symbols



Generate Passphrase

`GET /generate-passphrase`

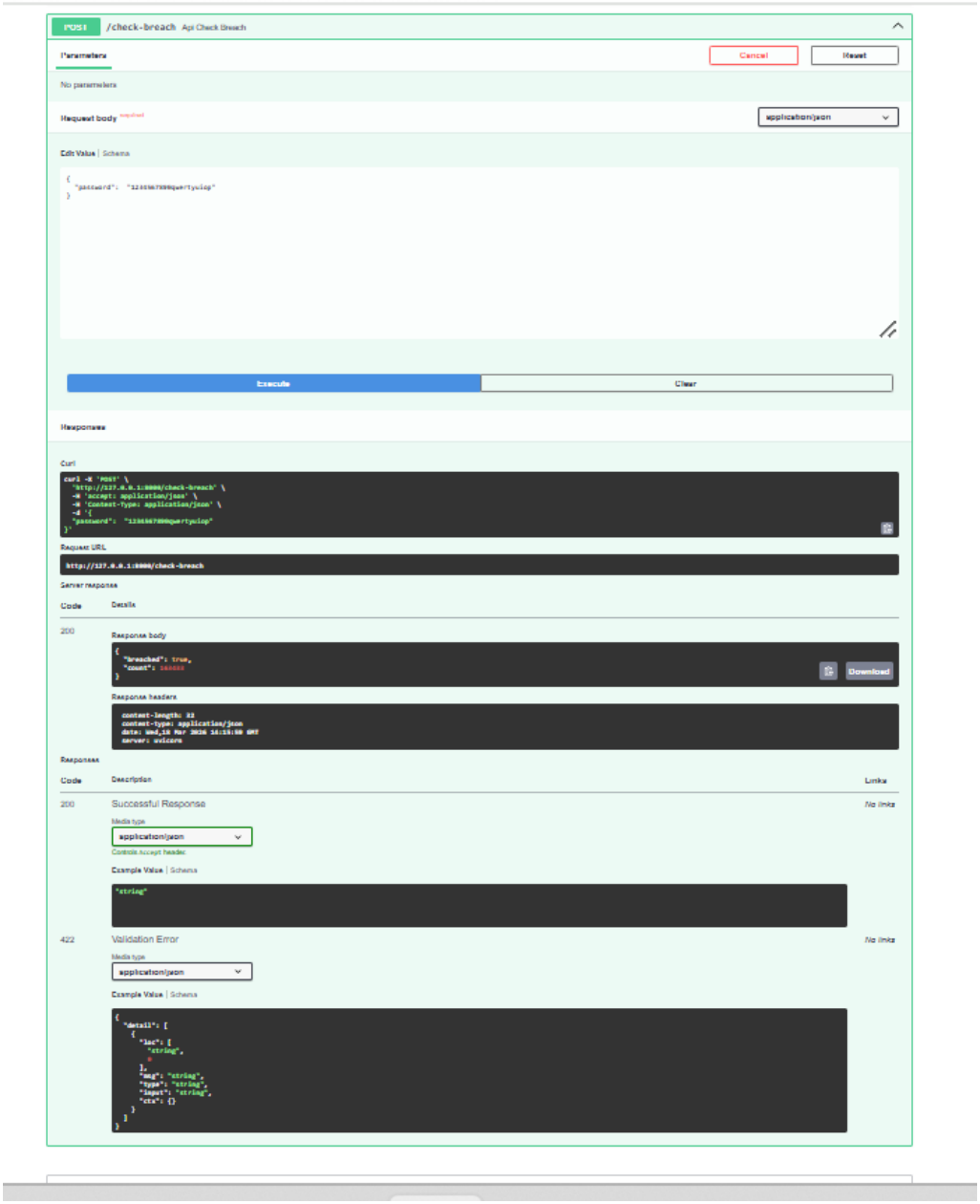
- Uses wordlist
- Easy to remember



Check Password Breach

`POST /check-breach`

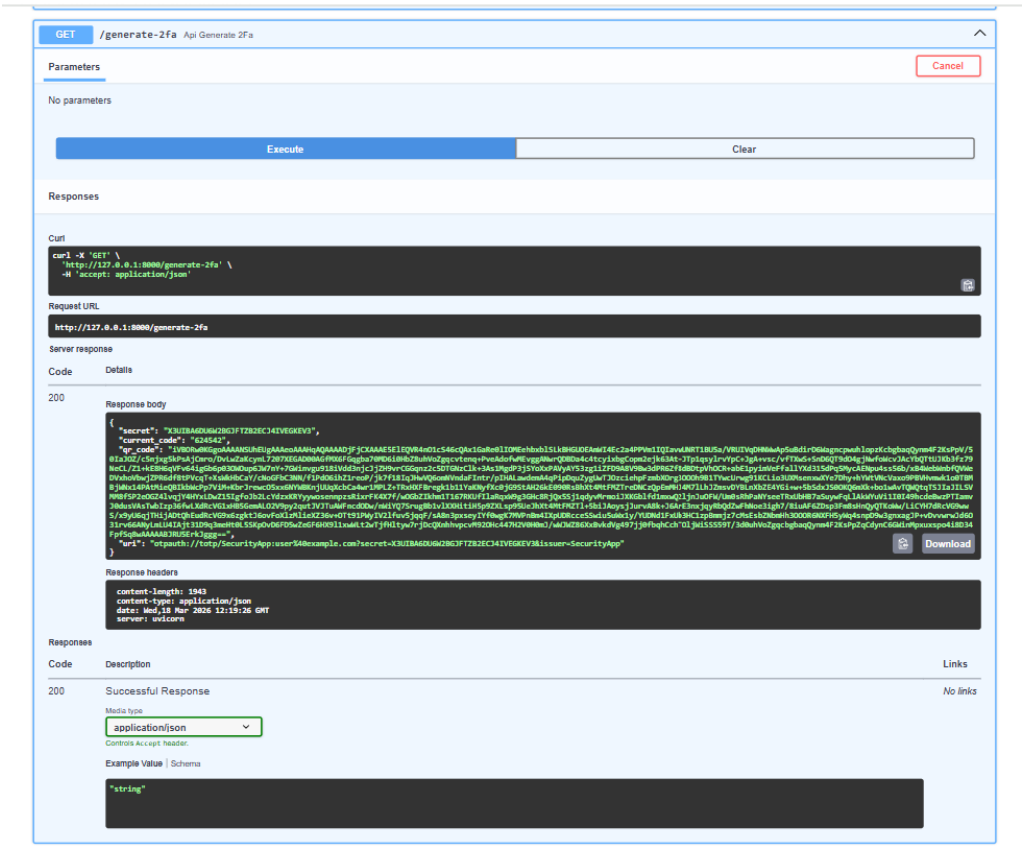
- Uses Have I Been Pwned API
- Checks if password is leaked



Generate 2FA

`GET /generate-2fa`

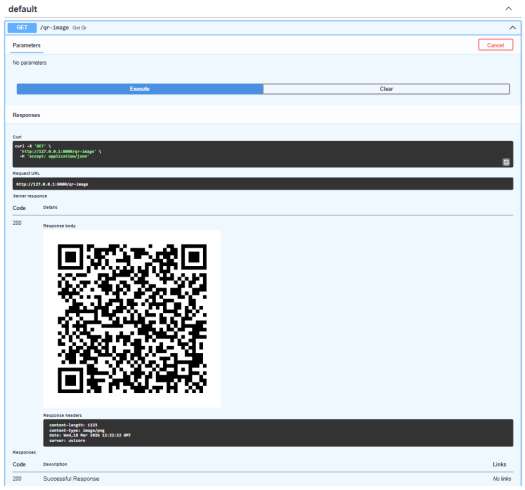
- Generates secret key
- QR Code
- 6-digit code every 30 seconds



QR Code

`GET /qr-image`

- Displays QR Code directly



Testing via Swagger

`http://127.0.0.1:8000/docs`

- Select endpoint
- Click Try it out
- Execute request

Security Notes

- Passwords are not stored in plain text
- SHA-1 used for API queries
- 2FA adds extra protection
- External API ensures real breach detection

Project Structure

```
project/

 security_module.py
 wordlist.txt
 images/
```

Conclusion

This project demonstrates modern security techniques and provides a strong foundation for secure system development.

Authors

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